National University of Computer and Emerging Sciences
Lahore Campus

AI (AI2002)

Sessional-I Exam

Date: Feb 24th 2026

Course Instructors:
AA, RM, BR, UA, UU
MN, SI, ST, HW

Total Time (Hr): 1
Total Marks: 27
Total Questions: 2

7+7=14

[redacted] Roll No

[redacted] Section

Student Name

Answer the questions in the provided space, no rough sheet to be attached.

CLO 1: *Demonstrate understanding of AI based strategies to systematically search for a solution*

Q1. A* Search for Autonomous Navigation [9 + 2 + 1 + 2 + 2 + 3 = 19 marks]

7

An autonomous delivery robot operates in a warehouse represented as a 2D grid.

- **Start state:** (0, 0)
- **Goal state:** (4, 3)

Movement Rules and Costs: From any cell, the robot can: Move horizontally or vertically → cost = 3 and Move diagonally → cost = 4

- **Constraints:** Blocked cells (obstacles) are (1,1), (2,1), (2,2), (0,3). The robot must stay within the bounds of x ε [0,4] and y ε [0,3].

Heuristic Function is the **Manhattan distance to the goal**:

$$h(n) = | x_{goal} - x_n | + | y_{goal} - y_n |$$

For node (2,0): $h(2,0) = | 4 - 2 | + | 3 - 0 | = 2 + 3 = 5$. Compute *g(n)*, *h(n)*, and *f(n)* for every generated node, where $h(n)$ is the estimate from that node to the goal. For example, when expanding (0,0) and generating (0,1), use heuristic $h(0,1)$.

(0, 3) OBSTACLE	(1, 3) 3	(2, 3) 2	(3, 3) 1	(4, 3) GOAL
(0, 2) 5	(1, 2) 4	(2, 2) OBSTACLE	(3, 2) 2	(4, 2) 1
(0, 1) 6	(1, 1) OBSTACLE	(2, 1) OBSTACLE	(3, 1) 3	(4, 1) 2
(0, 0) ROBOT	(1, 0) 6	(2, 0) 5	(3, 0) 4	(4, 0) 3

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a. **Search Tree Construction:** Generate the search tree expanded by the A* algorithm. For every node, clearly state the coordinate, the path cost, the calculated heuristic, and the total evaluation function. [7+2]

Tie-breaking rule: If two nodes have the same ***f(n)***, expand the one with the smaller x-coordinate. If x-coordinates are also equal, expand the one with the smaller y-coordinate.

b. Maintain both the frontier (open list) and the visited (closed list). [2]

c. When the goal is reached, write the final sequence of coordinates in the path and the total path cost. [1]

d. Is the provided heuristic *h(n)* admissible for this problem? How? [2]

e. Consider another heuristic for this problem.

$$h_2(n) = 4\,(\min(dx, dy)) + 3(|\,dx - dy\,|)$$

where $dx = |\,x_g - x\,|$ and $dy = |\,y_g - y\,|$.

Is h_2 admissible? How? [2]

f. From the given two heuristics (*h(n)* and *h₂(n)*), which one will make A* search more efficient for this problem? Justify. [3]

a) $\boxed{[h, g, f]}$

(0,0) [0,0,0]

[6,3,9] (0,1) — (1,0) [6,3,9]

[5,3,8] (0,2) — (1,2) [4,4,8] — (2,0) [5,3,8]

[4,3,7] (1,2) — (1,3) [3,4,7]

[3,3,6] (1,3) — (2,3) [2,4,6]

(2,3) [2,3,5]

,3,4] (3,3) — (3,2) [2,4,6]

(4,3) [0,3,3]

goal reached.

4

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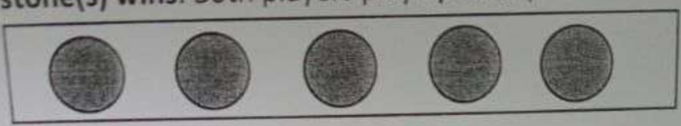
Q2. Minimax – Five-Stone Take-Away Game

A two-player, zero-sum, deterministic game starts with 5 stones in a single pile.
Players **MAX** and **MIN** alternate turns. On each turn, a player may remove **1 or 2** stones. The player who removes the **last stone(s) wins**. Both players play optimally. **[8 marks]**



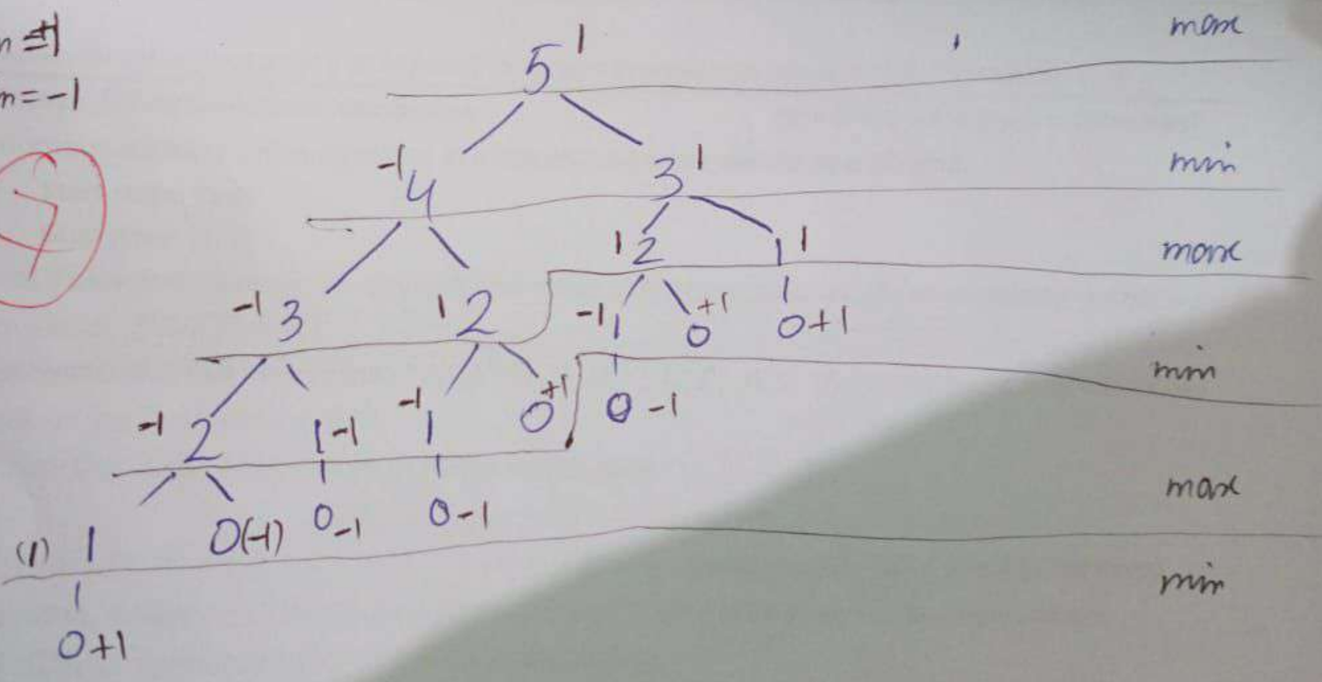
Tasks:

a. Build the complete game tree that will be traversed by the minimax algorithm to decide the move. Label levels alternately as MAX and MIN. [4]

b. Compute the value of each node in the game tree, propagate the values from the terminal nodes to the root, and determine the first move that will be selected by the player. [3 + 1]

let max win =+1
let min win=-1

7



(b) 1st move selected by player will be picking 1 stone.